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SCHOOL OF ENGINEERING
DEPARTMENT OF GEOMATICS ENGINEERING



A PROJECT REPORT ON
LAND SUITABILITY ASSESSMENT USING GIS AND MCDM
APPROACH FOR LARGE CARDAMOM PRODUCTION IN
LAMJUNG DISTRICT

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ABSTRACT

Large cardamom is a high-value spice crop of significant economic and cultural importance in the mid-hill regions of Nepal, particularly in Lamjung District. This study aims to evaluate the suitability of land for large cardamom cultivation in Lamjung by integrating key environmental and topographical factors, including elevation, slope, aspect, land use/land cover (LULC), temperature, precipitation, relative humidity, soil pH, and soil texture. Using high-resolution spatial datasets and applying a multi-criteria evaluation approach with GIS, each factor was classified into suitability classes based on the Land Suitability Evaluation Framework given by FAO (1976). The analysis reveals that 5.27% of the total area falls under not suitable and restricted zones, while 13.74% falls under marginally suitable zones, with 80.36% of the area being in moderately suitable zones, and 0.61% of the total area is covered by highly suitable zones for large cardamom production in Lamjung District. The soil texture is the highest influencing factor with 22.34% for large cardamom cultivation, while aspect is the lowest contributing factor with 2.45%.

This study contributes to the scientific understanding of the agro-ecological requirements and assists planners to make evidence-based decisions and provide practical recommendations on a sustainable future of cardamom in Lamjung District.

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LIST OF ABBREVIATIONS

AHP	Analytic Hierarchy Process
DEM	Digital Elevation Model
FAO	Food and Agriculture Organization
GIS	Geographic Information System
LULC	Land Use and Land Cover
MCDM	Multi-Criteria Decision Making
MCDA	Multi-Criteria Decision Analysis
PCM	Pairwise Comparison Matrix
CR	Consistency Ratio
CI	Consistency Index
RI	Random Index
KM	Kilometer
M	Meter

CHAPTER 1: INTRODUCTION

1.1 BACKGROUND

Cardamom, known as the Queen of Spices, is the world's oldest and is a leading crop that earns foreign exchange. Large cardamom (*Amomum subulatum* Roxb.), also known as Nepal cardamom, is one of Nepal's largest export crops (Bhattarai, 2016). It is one of the world's ancient species belonging to the family Zingiberaceae, popularly known as Alainchi in Nepali. It mainly grows in wild forest ecosystems and is also domesticated in the sub-Himalayan region, at altitudes ranging from 1000 to 2200m above mean sea level. It is a perennial herb and shade-loving plant which requires well-distributed rainfall of about 3000-5000 mm/year and is generally grown in tracts (Pandey et al., 2023). The height of this plant is 1.5 to 3.0, and on the upper part of the stem, there are leaves. The rhizomes are dull red, and the flower buds extend from the base of the rhizome. Spring is the flowering period of large cardamom flowers. It has short peduncle and buds covered with tight red bracts. The individual flowers remain open for three days or more. At the same time, new ones are opened. Flowering remains intact with the flowers for one month (Timsina & Pokhrel, 2018).

Nepal is the largest producer of cardamom in terms of both quality and quantity over the world. Cardamom is a perennial cash crop and gives the production for 10-15 years after planting. It can tolerate high humidity (>90 percent) and can be grown on swampy and riversides land with (>70%) soil moisture. Taplejung, Panchthar, Ilam, Shankhuwasabha, and Lamjung are the major cardamom-producing districts (Dawadi & Pant, 2024). Large cardamom prefers a moderate temperature range, generally between 10°C and 35°C. It grows well in forest loam soils, which are typically acidic (pH 5.5-6.5), and prefers Sandy loam to clay loam soils, which are sufficient with good drainage (Rahim et al., 2019).

The crop has a negligible opportunity cost of land, meaning it would not replace the land for other uses. So, this could provide additional opportunity to uplift the economic status of the people without any adverse effects on other farming. The suitability of climate and location, rainfall, temperature, export performance, tariff advantages, and availability of varieties of crops suitable to different geo-climatic conditions were other factors of the crop (Bhattarai, 2016). The cultivation of cardamom was recently introduced in Lamjung District, Nepal, and has been adopted due to economic and the crop being suited to the existing agro-ecological conditions. Many farmers have either dropped the traditional

crops or returned home after working abroad as labor migrants and instead engaged in commercial farming to generate higher incomes and improve their living standards (Acharya, 2019).

In the AHP, the decision-making process of complex problems is conducted by dividing the problem into issues, which may be divided further to form a simple and comprehensible hierarchical structure developed by Saaty in 1980, which is considered a mathematical approach to MCDM (Bouamrane et al., 2022). Various research have been conducted using a geographic information system (GIS), a processing tool capable of collecting, creating, storing, and managing various types of spatial data. GIS can be used to integrate many kinds of data or variables and build new sets of information (Harist et al., 2018).

1.2 PROBLEM STATEMENT

Large cardamom (*Amomum subulatum*) is one of Nepal's most valuable high-altitude cash crops, contributing significantly to the rural economy, especially in the eastern hills. However, despite its favorable climatic and ecological diversity, districts like Lamjung have not yet fully realized the crop's potential. While some neighboring regions with similar geographic conditions have adopted cardamom farming extensively, Lamjung remains underutilized due to the lack of spatially informed agricultural planning, insufficient awareness among farmers, and inadequate identification of suitable cultivation zones based on scientific criteria.

In this context, there is a critical need for a scientific approach to identify the most suitable lands for large cardamom cultivation to optimize resource use, improve productivity, and support sustainable farming. By integrating Geographic Information Systems (GIS) with Multi-Criteria Decision Making (MCDM) techniques such as the Analytical Hierarchy Process (AHP), this project aims to assess and map areas optimal for large cardamom cultivation. Factors such as elevation, slope, aspect, precipitation, temperature, humidity, soil pH, soil texture, and land use/land cover (LULC) are considered to model the suitability zones. The outcome will provide a strategic guideline for stakeholders to promote cardamom farming in appropriate areas, ensuring sustainable land use and improved agricultural productivity.

1.3 OBJECTIVE

1.3.1 Primary objective

The primary objective of the project was to evaluate the land suitability for large cardamom (*Amomum subulatum*) cultivation in Lamjung district using GIS and MCDM approach.

1.3.2 Secondary objectives

The secondary objective of the project was to:

- To identify and map key biophysical factors (e.g., elevation, slope, aspect) influencing cardamom cultivation.
- To reclassify thematic layers into suitability classes using previous research and agro-climatic thresholds from FAO (1976).
- To perform a weighted overlay analysis and classify the land into different suitability zones.

1.4 SCOPE OF THE PROJECT

The scope of the project was to evaluate the land suitability for large cardamom (*Amomum subulatum*) cultivation in Lamjung district using GIS and MCDM approach. Specifically, this project intended to:

- Identify the key factors influencing land suitability for large cardamom cultivation such as slope, elevation, LULC, Temperature and soil texture etc..
- Develop hierarchical decision models using AHP approach to evaluate the relative importance of the identified large cardamom land suitability factors.
- Apply the AHP with numerous GIS tools to generate suitability map for cardamom production with suitability classes like high, moderate, marginal, not suitable and restricted.
- Calculate the area of the different classes of land using field calculator to analyze the maximum and least zones of suitability.
- Validate the suitable map obtained from the analysis using the prime location where the cardamom cultivation is being practiced in recent times.

CHAPTER 2: LITERATURE REVIEW

In the developmental context, a lot of technology has gone through major breakthrough phase with tremendous advancement. This technology includes GIS and other survey techniques, and as the approach of MCDA/MCDM, as well as the integration of both (Malczewski & Rinner, 2015). Land suitability analysis is a critical step for optimizing agricultural production, especially for high-value crops like large cardamom (*Amomum subulatum*). The integration of Geographic Information Systems (GIS) and multi-criteria decision-making tools such as the Analytic Hierarchy Process (AHP) has become a widely accepted approach for evaluating land suitability by incorporating multiple environmental and socio-economic factors (Mohammed Abdule & Kejela Woyesa, 2023). These tools allow for systematic weighting and spatial analysis of criteria such as elevation, slope, soil properties, climate variables, and land use, providing a robust framework for identifying suitable cultivation areas.

Several studies have demonstrated the effectiveness of GIS and AHP where a study conducted by (Baniya et al., 2009) on physical land assessment for the large cardamom cultivation in hills of Kathmandu Valley showed that more than one-third of total land areas above 1800m above sea level are suitable. However, topographic constraints (steep slope, length, orientation, etc.) restricts suitability ranking. It was found to be grown well in north facing slope of more than 5° and thirteen percent of the total land area under study having gentle slope (3–8°) with loamy soil are classified as highly suitable (S1).

Specific to cardamom, numerous studies have applied FAO land evaluation frameworks combined with GIS and field data to classify land suitability. (Faisal et al., 2025) evaluated cardamom cultivation potential in Indonesia by assessing rooting media, nutrient retention, slope, and climate factors, categorizing land into highly suitable to not suitable classes. Their findings emphasized the importance of well-drained soils, moderate slopes, and adequate rainfall for optimal cardamom growth. Similarly, (Shivaprasad et al., 2001) conducted a soil-site suitability evaluation for cardamom in India, identifying that well-distributed rainfall (1500–2000 mm), soil depth (>80 cm), and elevation (900–1200 m) are key factors for high suitability. They also suggested that marginally suitable lands could be improved through management practices such as mulching and shade regulation.

In the Himalayan context, a GIS-based land suitability model for *Amomum compactum* in West Sumatra used rainfall, soil type, and elevation as primary variables, producing a suitability map that identified 53% of the area as suitable for cardamom cultivation (Harist et al., 2018). This approach underscores the adaptability of GIS-AHP methods to different geographic and climatic contexts for cardamom production.

Despite the growing body of research, challenges remain in large cardamom production, particularly in Nepal's Lamjung district, where diseases like chirkey and rhizome rot, poor seedling quality, inadequate irrigation, and socio-economic factors limit productivity (Dawadi & Pant, 2024).

2.1 CARDAMOM PRODUCTION INFLUENCING FACTORS

The following influencing factors for land suitability of cardamom production were considered during our project through meaningful literature reviews:

a) Elevation

Elevation significantly influences the microclimate, temperature, and humidity, which are critical for large cardamom growth. Optimal elevation ranges ensure suitable temperature and moisture conditions, minimizing disease risks and maximizing yield. Studies in Himalayan regions show that large cardamom thrives best between 1200 and 2000 meters (Faisal et al., 2025).

b) Slope

Slope affects soil drainage, erosion risk, and mechanization feasibility. Moderate slopes provide good drainage essential for cardamom, which is sensitive to waterlogging. Excessively steep slopes increase erosion and reduce suitability (Shivaprasad et al., 2001).

c) Aspect

Aspect determines solar radiation exposure and microclimatic conditions such as moisture retention. North and northeast-facing slopes are preferred for large cardamom due to cooler temperatures and higher humidity, which reduce evapotranspiration stress (Dawadi & Pant, 2024).

d) Land Use/ Land Cover (LULC)

LULC indicates existing vegetation and land management practices. Large cardamom requires shaded environments often provided by forest or agroforestry systems. Areas dominated by trees or agroforestry have higher suitability compared to barren or built-up lands.(Dhungana et al., 2024).

e) Temperature

Temperature controls physiological processes and disease development in cardamom plants. Optimal growth occurs between 10°C and 22°C; temperatures outside this range stress plants and reduce (International Journal of Environmental & Agricultural Biology, 2025; Faisal et al., 2025).

f) Precipitation

Adequate and well-distributed rainfall (2000–3500 mm annually) is essential for maintaining soil moisture and humidity. Insufficient or erratic rainfall can lead to drought stress, while excessive rainfall may cause waterlogging and disease (Dhungana et al., 2024).

g) Relative Humidity

High relative humidity (above 65%) favors large cardamom growth by reducing transpiration stress and supporting flowering and fruit set. Low humidity increases vulnerability to pests and diseases (Dawadi & Pant, 2024; Faisal et al., 2025).

h) Soil pH

Slightly acidic soils with pH between 5.2 and 6.5 optimize nutrient availability and microbial activity, enhancing cardamom growth. Soils outside this range can limit nutrient uptake and increase susceptibility to diseases (Shivaprasad et al., 2001; FAO, 2019).

i) Soil Texture

Well-drained loamy soils rich in organic matter provide ideal root aeration and moisture retention for cardamom. Heavy clay soils cause waterlogging, while sandy soils may not retain sufficient moisture (AIP Publishing 2018; Rahim et al., 2019).

CHAPTER 3: METHODOLOGY

3.1 CONCEPTUAL FRAMEWORK

The proposed project uses GIS integrated with AHP to identify the most suitable areas to cultivate cardamom by combining Geographic Information System (GIS) with the Analytic Hierarchy Process (AHP). The study incorporates the criteria like elevation, slope, aspect, soil PH, soil texture, rainfall, temperature, humidity, and land use/land cover. The team selected the most significant factors after analyzing recommendations of supervisors with reading previous research and prioritized them through AHP. The comparison of each factor with all other factors was done to develop weights that indicate the extent to which each factor influences the growth of cardamom. The weights were typically highest on elevation, slope and soil type.

The team verified the consistency ratio of AHP matrix. Using these weights, GIS produced a new map in which all the layers were reclassified into different suitability levels. Then weighted overlay analysis was performed to obtain the final suitability map. This map classifies Lamjung district into zones: highly suitable, moderately suitable, marginally suitable, not suitable and restricted zones of cardamom cultivation. The research recommends that cardamom should be promoted in the most appropriate regions, agroforestry should be encouraged in the forest-buffer regions, and the planners, farmers, and policymakers should have significant information to develop sustainable agricultural practices in the Lamjung District

3.2 STUDY AREA

Lamjung District is located in Gandaki Province of Nepal, measuring 1,692 square kilometers in area, and has an altitude between 600 m and 8016 m. It spans from tropical to trans-Himalayan geo-ecological zones and is known for its diverse geography and mixed ethnic population, including a significant Gurung community. The district headquarters is Besisahar, situated 175 km west of Kathmandu. The approximate latitude and longitude range of Lamjung District in Nepal is from 28° 7'N to 28° 21'N and from 84° 14'E to 84° 35'E

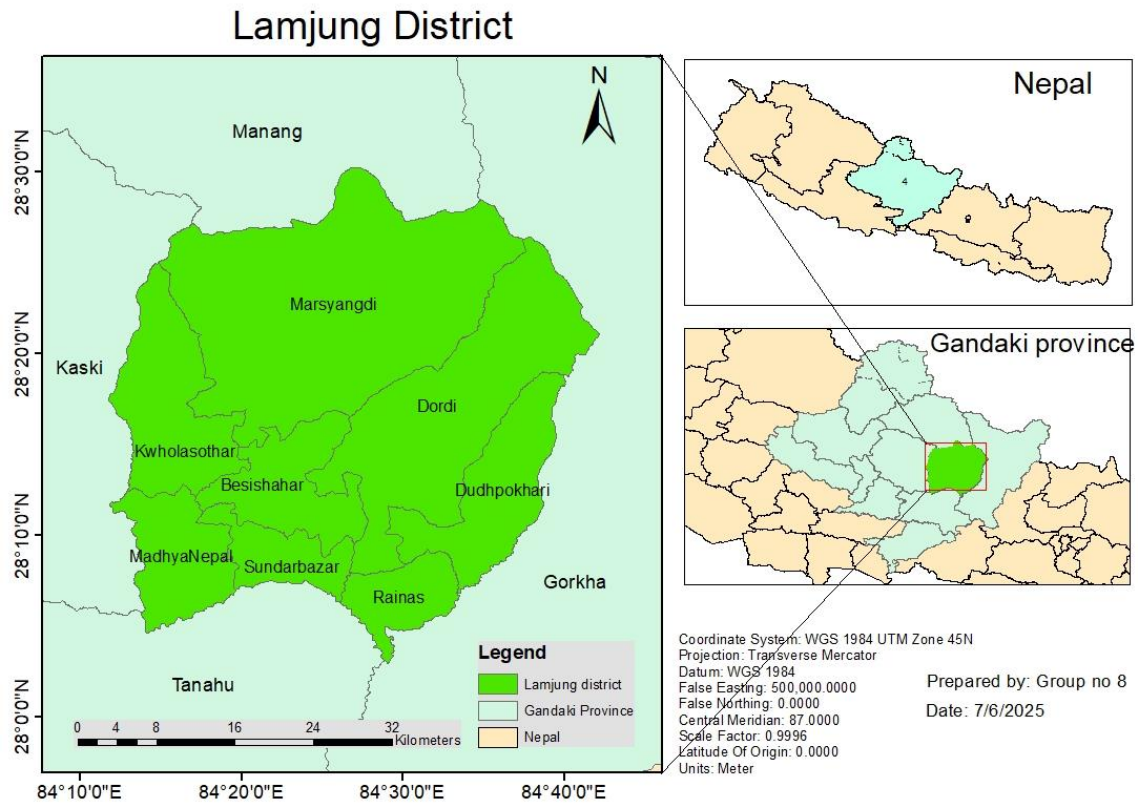


Figure 1: Study Area of Lamjung District

3.3 SOFTWARE USED

➤ ArcGIS

ArcGIS is a widely used GIS software platform that enables users to perform spatial data analysis, mapping, and suitability modeling. In suitability analysis, ArcGIS allows us to combine multiple criteria such as elevation, slope, aspect, soil PH, humidity, precipitation, , temperature, soil texture, and LULC by assigning weights to each factor and calculating a suitability score for every location.

➤ Microsoft Excel

Microsoft Excel was used to create horizontal bar charts for each land suitability criterion. In Excel, data for criteria such as elevation, slope, aspect, precipitation, temperature, soil pH, soil texture, LULC, and humidity were entered in columns. Additionally, the consistency ratio check using various AHP calculations was performed in Excel.

3.4 DATA USED

Table 1: Data Used

S.N.	DATA TYPE	SOURCE	DATA RESOLUTION	RESULT
1.	DEM	ALOS PALSAR	Raster (12.5m)	Slope, Aspect, Elevation
2.	Soil Ph	ISRIC World Soil Information	Raster (250m)	Acidity in soil
3.	Temperature	Climatology Lab Terraclimate	Raster(4km)	precipitation Amount
4.	Precipitation	CHIRPS Climate Hazards Center	Raster(5km)	Temperature range
5.	Humidity	Climatology Lab Terraclimate	Raster(4km)	Moisture content in air
6.	LULC	ESRI land cover	Raster(10m)	Land use and land cover
7.	Soil Texture	ISRIC World Soil Information	Raster (250m)	Defines soil physical properties

3.5 METHODOLOGICAL WORKFLOW

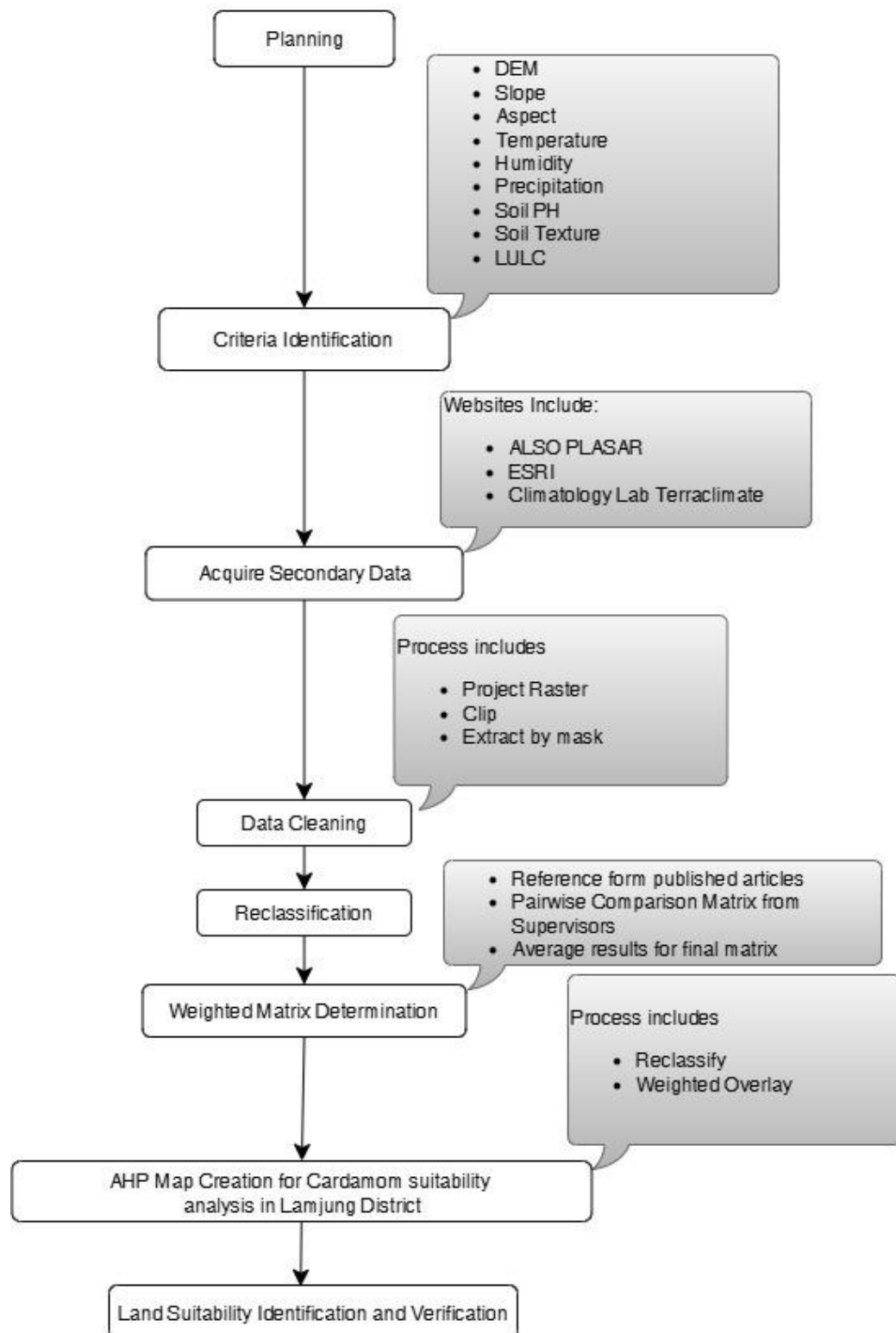


Figure 2: Workflow

a) Planning

In planning, a suitable study area was chosen where the secondary data is accessible, with the best possible high resolution, and had some prior practice of cardamom cultivation, so that it would assist us in the validation of our project. The list of possible secondary sources was determined. The available topographical map and the report of the current practice of large cardamom production on different local levels of Lamjung District were studied. The different research performed on land suitability for large cardamom production with an integrated approach of AHP and GIS was studied.

b) Criteria Identification

The first step in the land suitability analysis process involves identifying the criteria that significantly influence the production of large cardamom. These criteria are selected based on a comprehensive review of scientific literature, expert consultations, and the known environmental requirements of the crop. For cardamom cultivation, important biophysical and climatic factors include elevation, slope, aspect, temperature, humidity, precipitation, soil pH, soil texture, and land use/land cover (LULC). By selecting these criteria, the study aims to incorporate all the essential environmental and land characteristics that contribute to successful cardamom growth in the region.

c) Acquiring Secondary Data

After identifying the relevant criteria, the next step was to acquire spatial data representing each of them. These data are obtained from various reliable global and national sources in raster format. Digital Elevation Models (DEMs) from sources like ALOS PALSAR provide information on elevation, slope, and aspect. Climatic variables such as precipitation, temperature, and humidity are collected from datasets such as TerraClimate and CHIRPS, which offer monthly and annual climate records at varying spatial resolutions. Soil-related data, such as pH and texture, are accessed from ISRIC SoilGrids while land use data is typically derived from ESA WorldCover or classified Sentinel-2 imagery.

d) Data Cleaning

Once the data has been collected, it must undergo a data cleaning process to ensure consistency and compatibility among the various layers. This involves several preprocessing steps, including projection standardization, masking, and resolution matching. All raster layers are reprojected to a common coordinate system (typically WGS 1984) to maintain spatial consistency. The study area, Lamjung district, is then used to clip or extract only the relevant portions of the datasets using tools like "Clip Raster," "Extract by Mask," or "Project Raster" in GIS software.

e) Reclassification

After cleaning and standardizing the data, each raster layer is reclassified into a set of standardized suitability classes. This reclassification simplifies complex continuous variables that reflect the favorability of each criterion for cardamom cultivation. Typically, the values are categorized into five classes, i.e., 5-Highly Suitable, 4-Moderately Suitable, 3-Marginally Suitable, 2-Not Suitable, and 1-Restricted. Based on the FAO (1976) guidelines and GIS-MCDM research in Nepal, the following ranges of criteria and sub-criteria are established to classify suitability.

Table 2: Effect and Criteria Rating

S.N.	Suitability Zones	Criteria Rating
1.	Highly suitable	5
2.	Moderately suitable	4
3.	Marginally suitable	3
4.	Not suitable	2
5.	Restricted	1

I) Elevation

Elevation is the height of the land above the average sea level. Cardamom should also be taken into consideration because it thrives in an area that is between 600 and 2,000 meters above sea level. Altitude affects temperature, rainfall, and other climatic factors that determine crop growth and development.

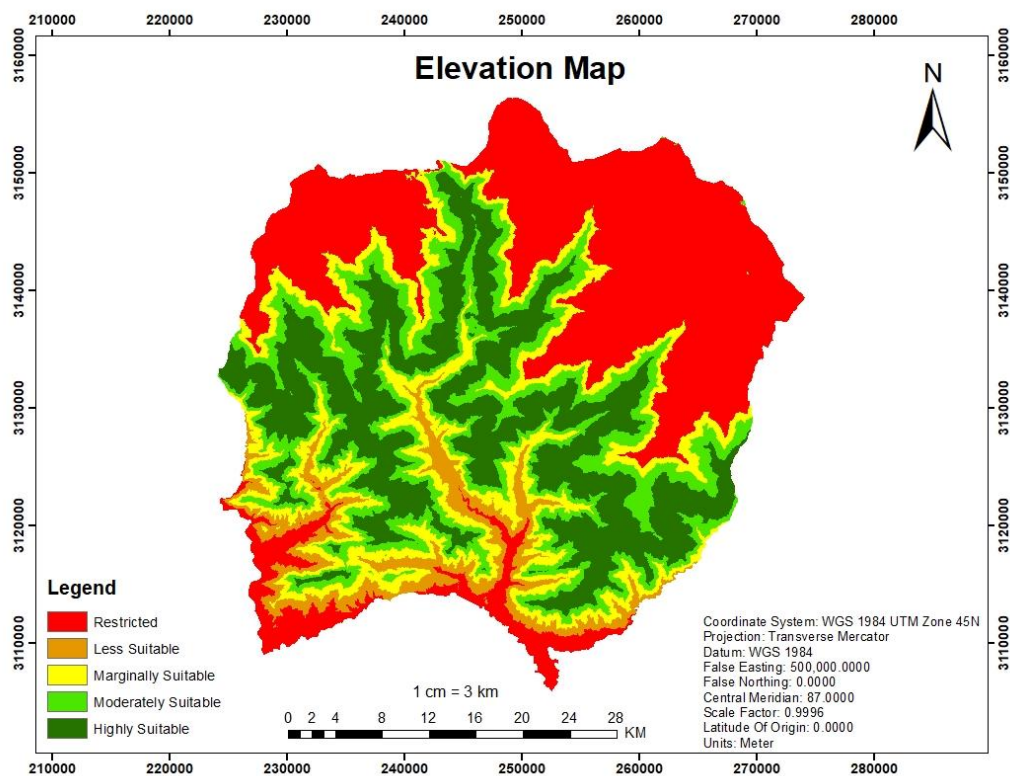


Figure 3: Elevation Map

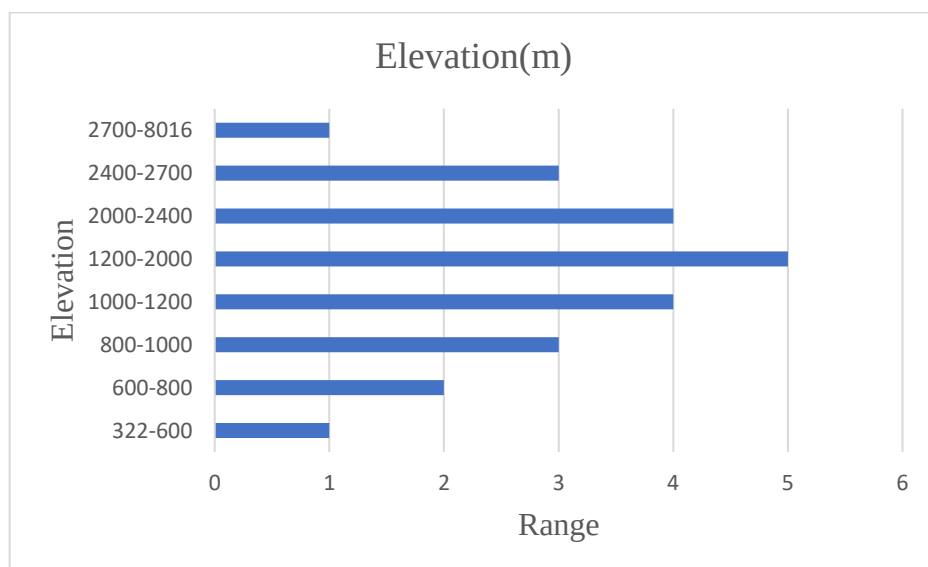


Figure 4: Elevation Chart

II) Slope

Slope is the degree of steepness or flatness of a land surface expressed in degrees or percent. Slope influences the drainage, erosion and the feasibility of agriculture. Cardamom is most suited to grow in gentle slopes or moderately steep slopes, which are less than 25 degrees. In the case of extremely steep slopes, the erosion is greater and soil stability is reduced.

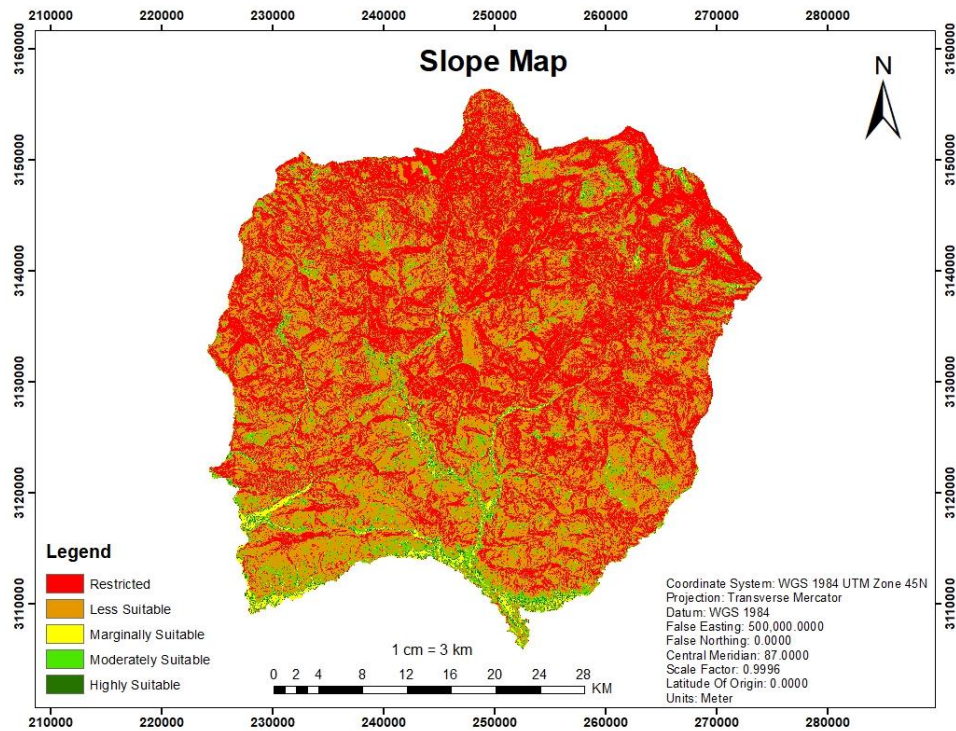


Figure 3: Slope Map

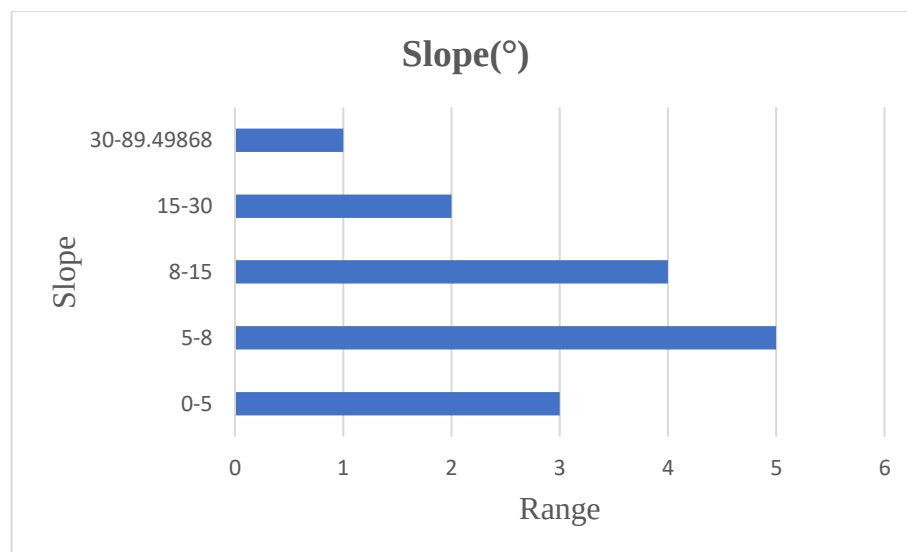


Figure 4: Slope Chart

III) Aspect

Aspect is the direction a slope faces in the compass. It influences the amount of sun that reaches there, and this alters the microclimate, which influences the soil moisture and temperature. In the case of cardamom, the north and east slopes are normally preferred due to their ability to retain moisture and provide a good growing environment.

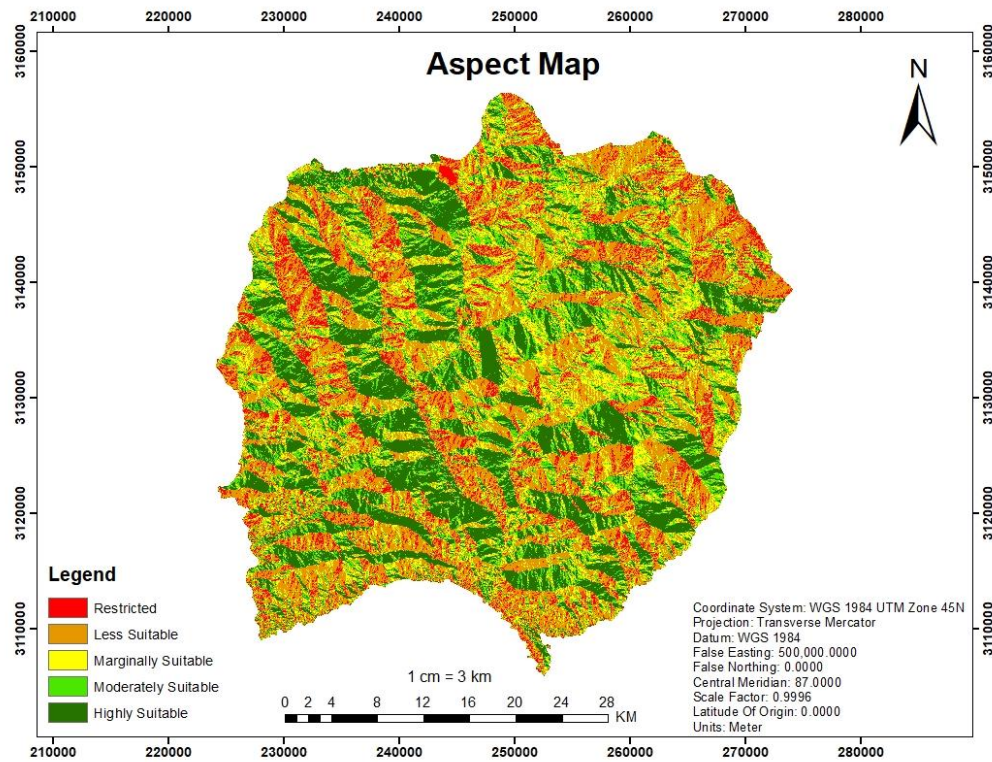


Figure 5: Aspect Map

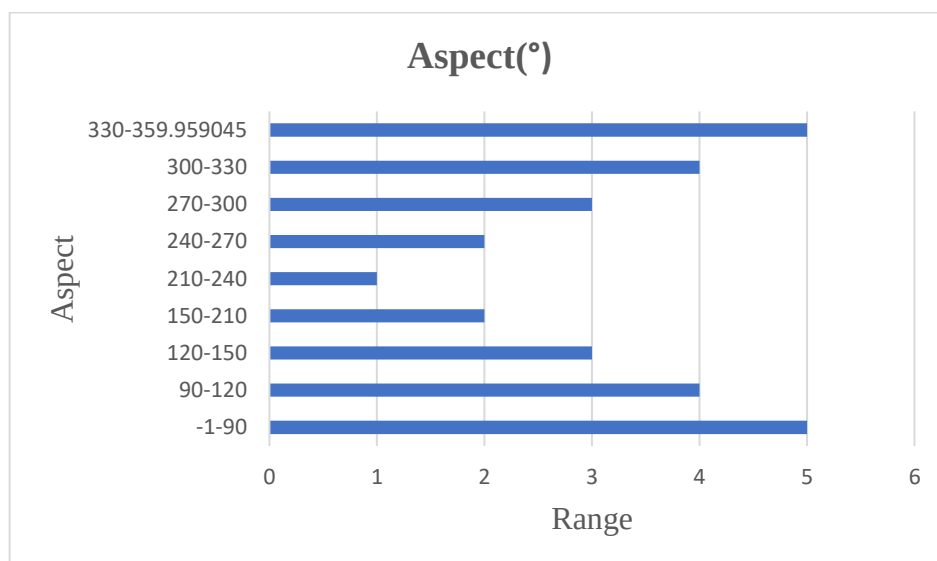


Figure 6: Aspect Chart

IV) Precipitation

Precipitation refers to the total rainfall and other forms of water, which is normally quantified in millimeters per annum. Cardamom requires rainfall that is well distributed throughout the year, approximately 1,500 to 2,500 mm. This quantity of rain maintains the soil wet and allows plants to remain healthy.

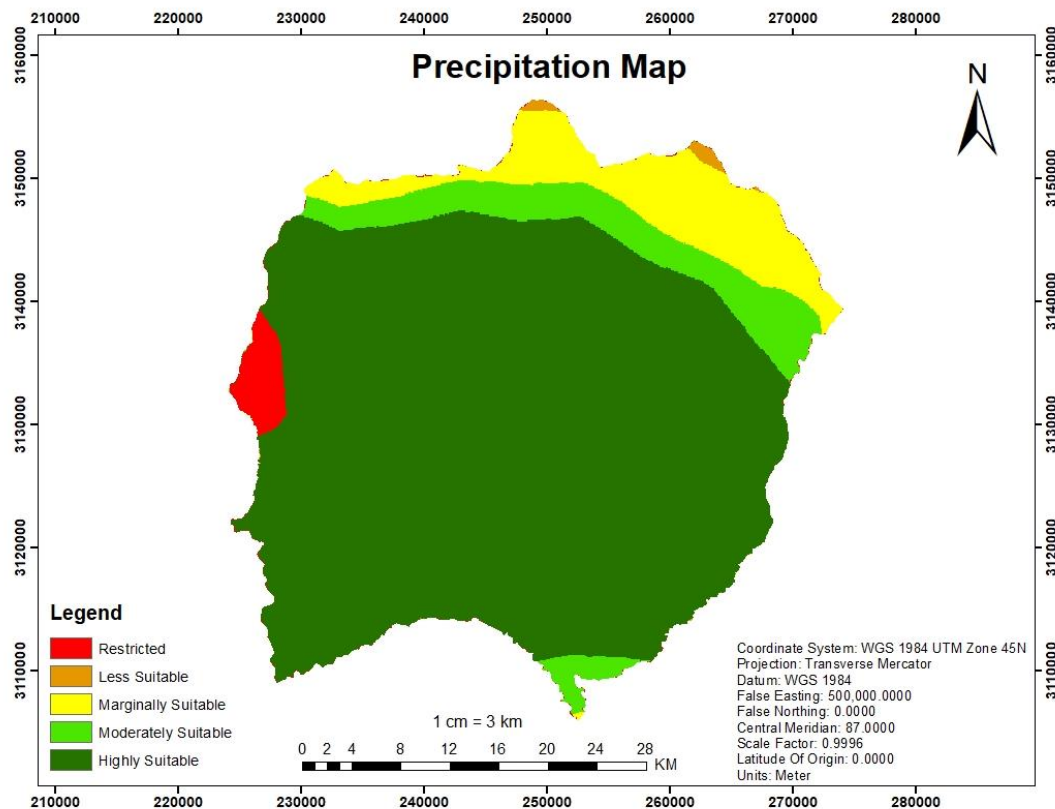


Figure 7: Precipitation Map

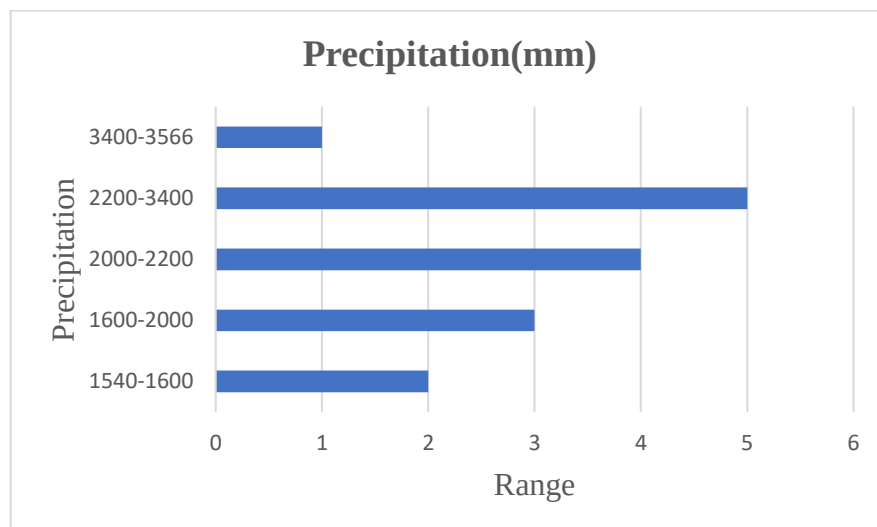


Figure 8: Precipitation Map

V) Temperature

One of the primary weather factors which affect the way plants grow and operate is temperature. Cardamom thrives well in moderate temperatures of 8-20°C. The plants can be stressed by hotter or colder weather and reduce yields.

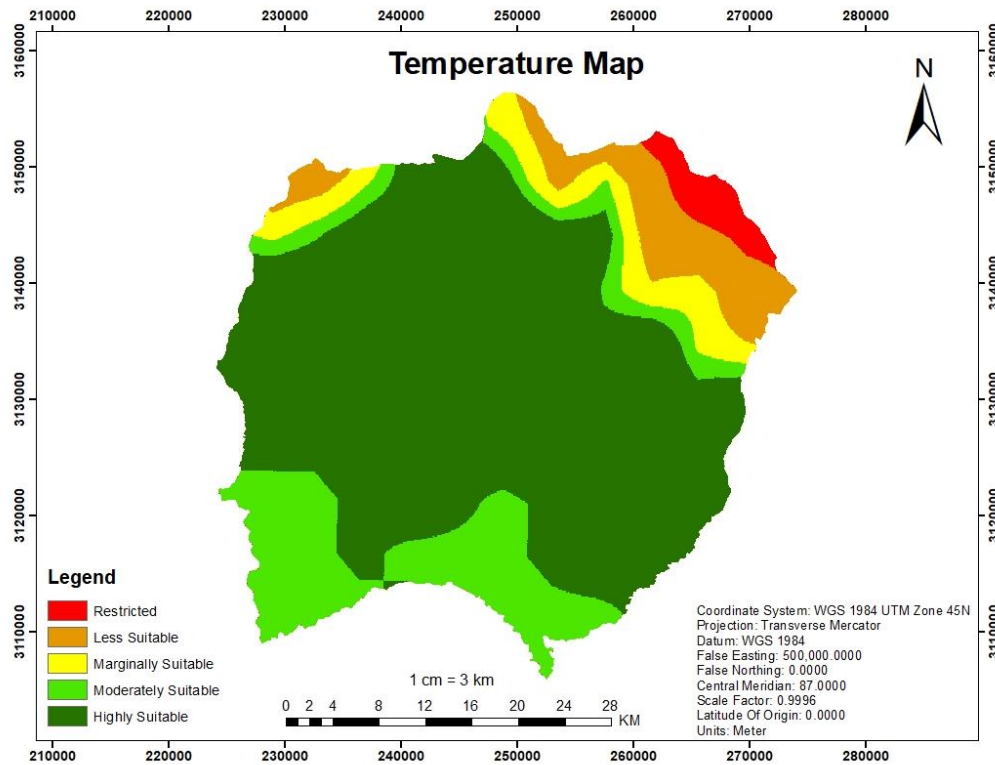


Figure 9: Temperature Map

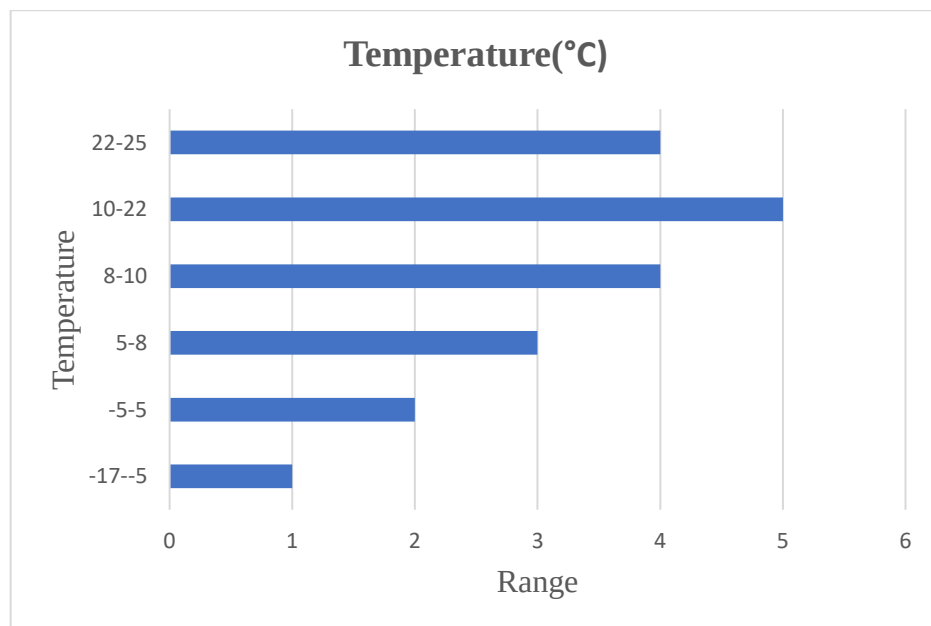


Figure 10: Temperature Chart

VI) Soil PH

The PH of the soil informs us of the acidity or alkalinity of the soil. It ranges between 0 (very acid) to 14 (very alkaline) and 7 is neutral. Cardamom prefers soils that are slightly acidic ranging between 5.5 and 6.5. Nutrients become difficult to access when the pH is beyond this range and crop productivity may decline.

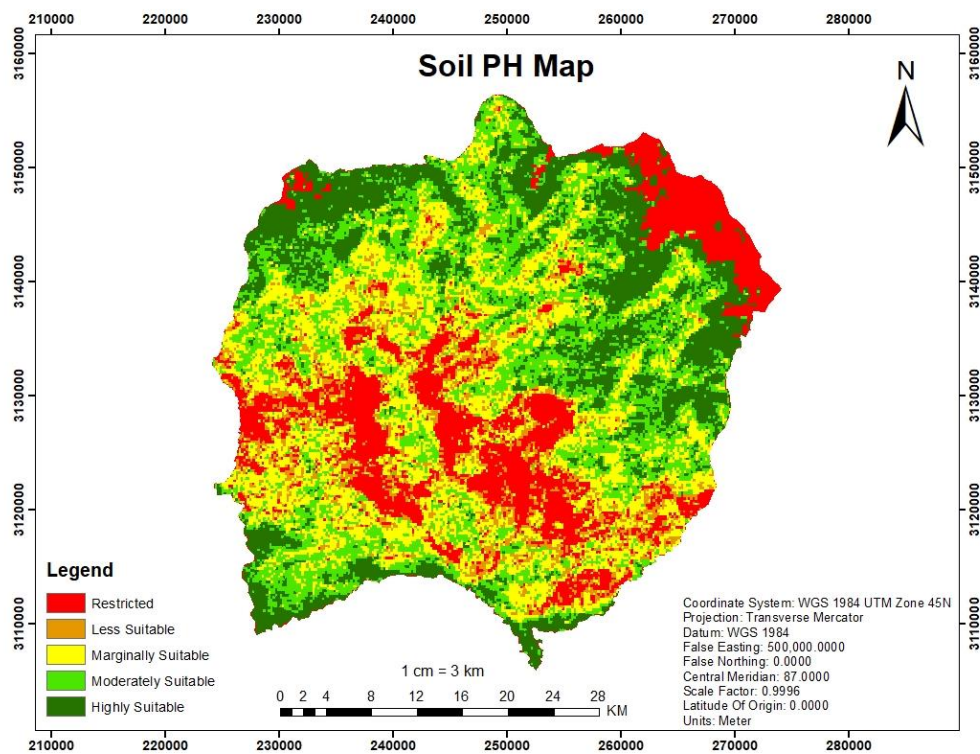


Figure 11: Soil PH Map

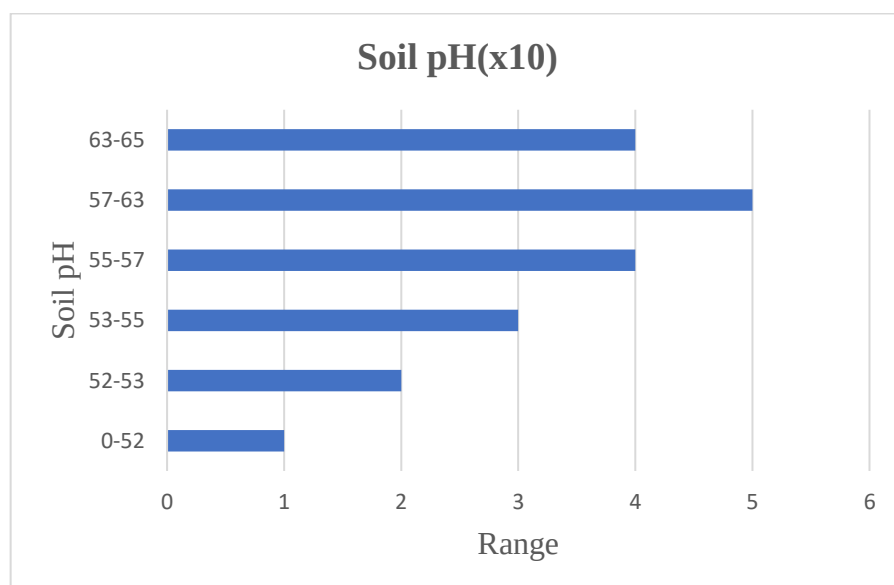


Figure 12: Soil pH Map

VII) Humidity

Humidity refers to the amount of water vapor present in the air. High humidity levels are beneficial for cardamom, as they help maintain leaf turgor, reduce water loss, and create a favorable environment for this shade-loving crop.

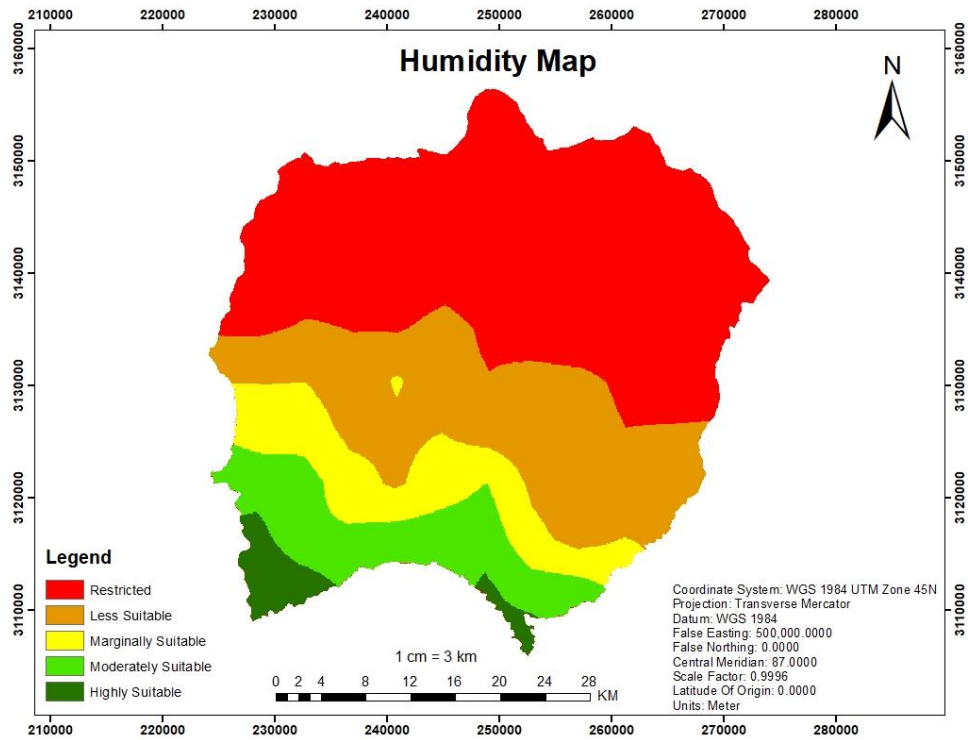


Figure 13: Humidity Map

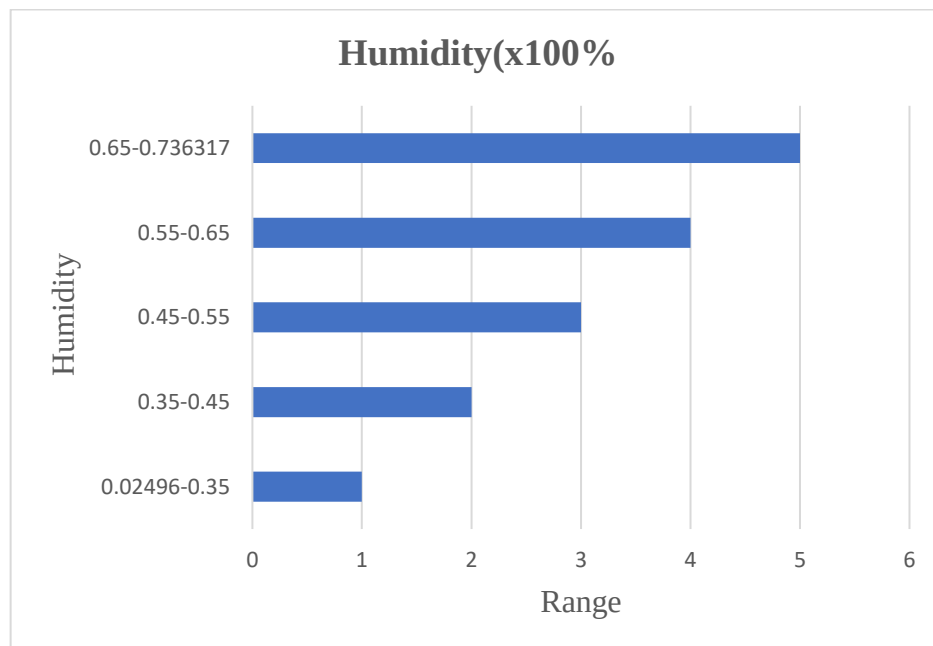


Figure 14: Humidity Chart

VIII) Land use and land cover (LULC)

(LULC) is another important data set that indicates the location of various types of land. It monitors vegetation, cropland, forest and other land types. The information will assist us in identifying areas already under cultivation, forest cover that can be used as shade (cardamom is sensitive to shade) and where crops or forests can be extended or conserved.

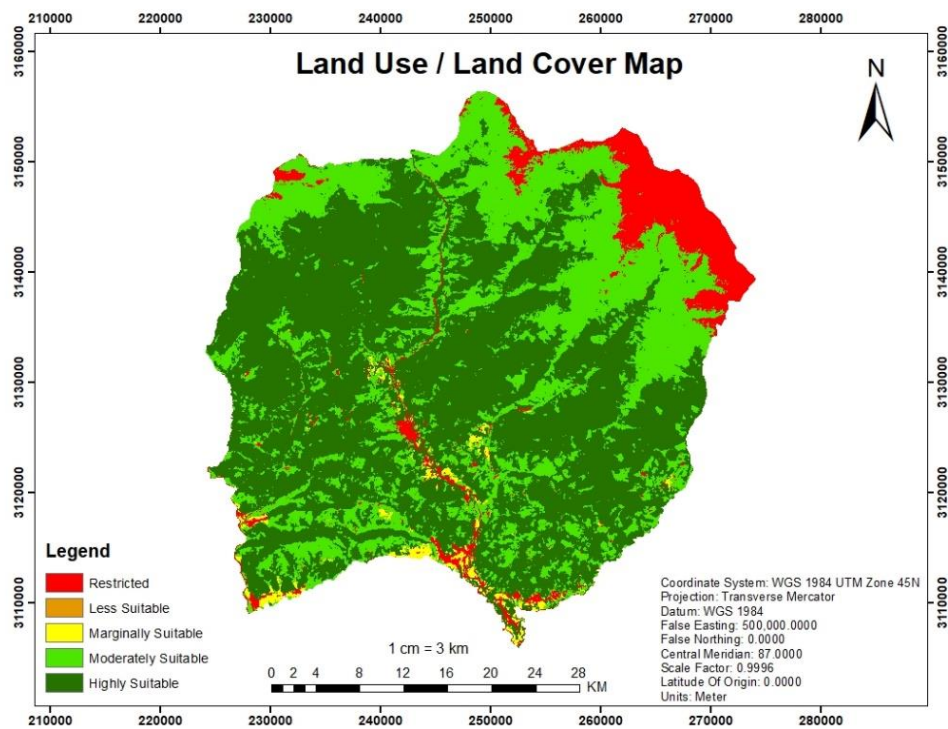


Figure 15: LULC Map

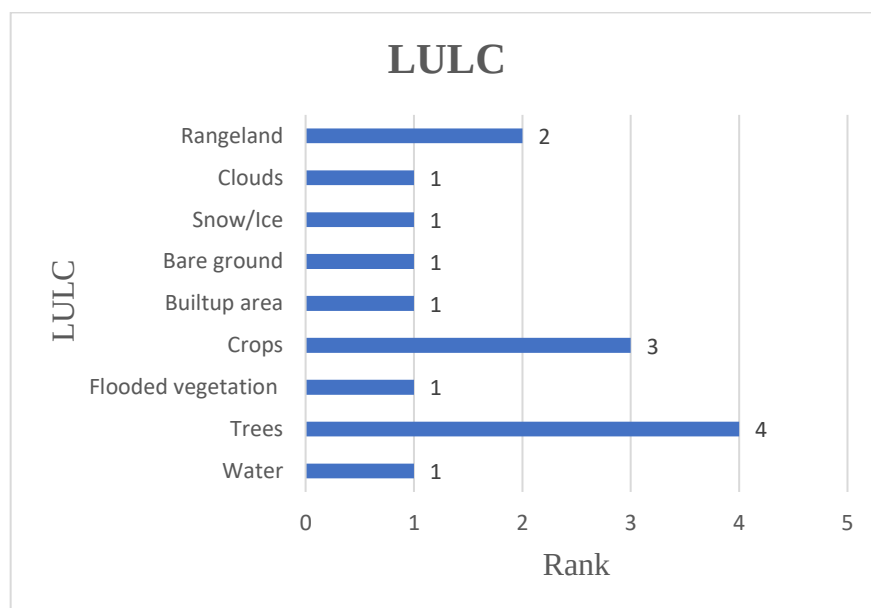


Figure 16: LULC Chart

IX) Soil Texture

The texture of the soil is the combination of the silt, clay, and sand particles that constitute the soil. Sandy loam to loamy soil texture is best suited to the cultivation of cardamom. They are well-drained soils but have sufficient moisture, which is what cardamom roots require. Nepal and Indian research indicate that cardamom is most commonly cultivated in soils that are sandy loam, loamy sand, silty loam, or clay loam, although the finest performance is on deep, well-drained loamy soils with high organic matter content.

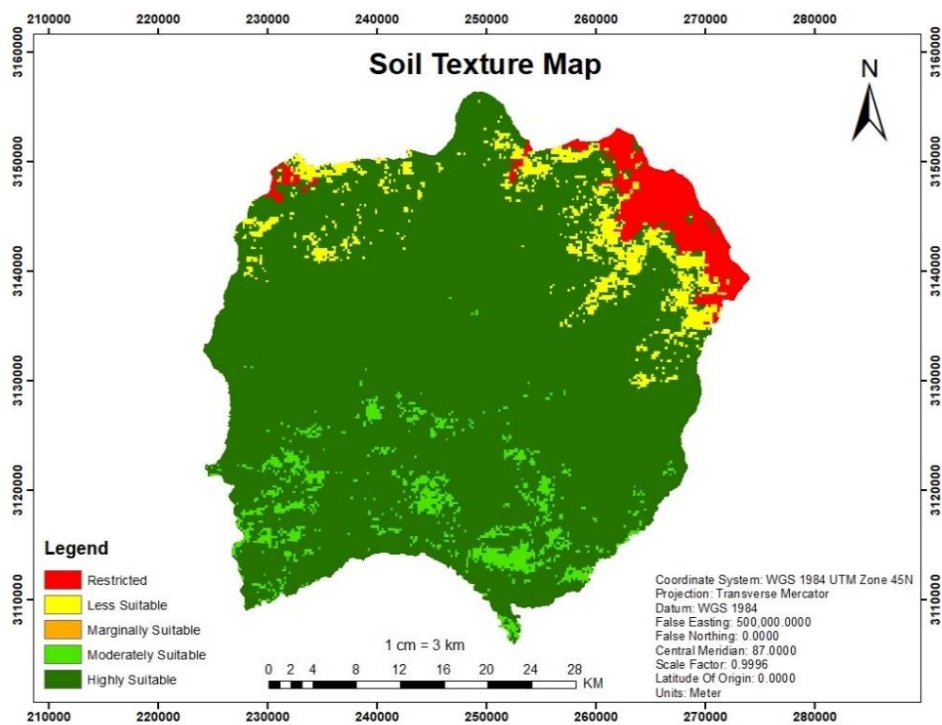


Figure 17: Soil Texture Map

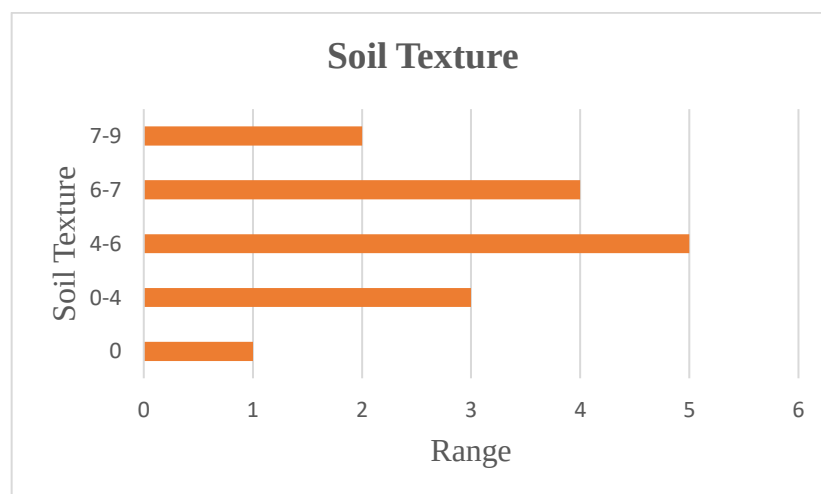


Figure 18: Soil Texture Chart

f) Weighted Matrix Determination

The Analytic Hierarchy Process (AHP) is a powerful decision-support method used in Multi-Criteria Decision Making (MCDM) that helps prioritize and assign weights to various factors involved in complex problems. In this study, AHP was applied to evaluate land suitability for cardamom cultivation by combining expert knowledge with geospatial data. This approach enabled quantifying the relative influence of biophysical factors and the generating a scientifically sound suitability map for Lamjung District. The process follows as:

Step 1: Defining Objective & Criterion

The major goal or the problem statement, and the criterion for assessment were identified in the first step. The sub-criteria of the criterion were also identified. **Figure 19** shows the hierarchical modelling in AHP between goals, criteria, sub-criteria, and alternatives.

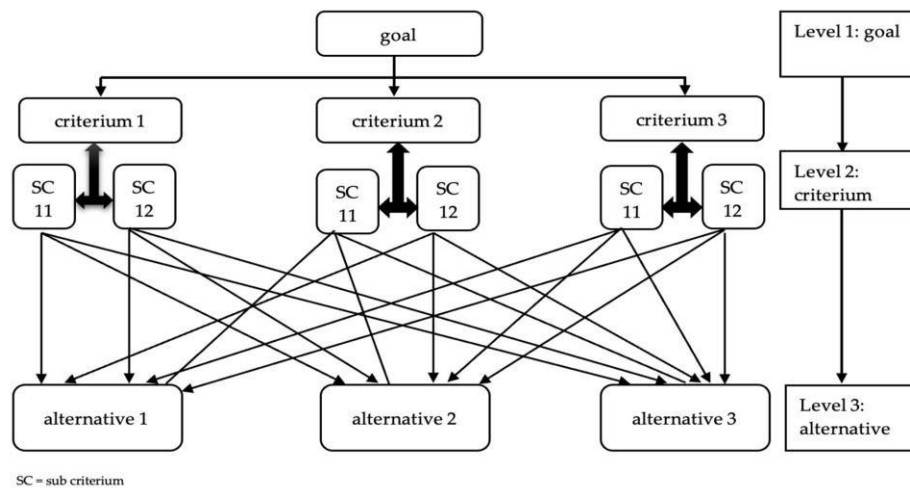


Figure 19: AHP Modelling

Step 2: Creating Pairwise Comparison Matrix

Table 3 provides insight into how the pairwise comparison matrix is done and created

Table 3: Pairwise Comparison Matrix

	Element 1	Element 2	...	Element n
Element 1	1	a_{12}	...	a_{1N}
Element 2	$1/a_{12}$	1		a_{2N}
....	...	...		...
Element N	$1/a_{1N}$	$1/a_{2N}$	...	1

Step 3: Individual Comparison of Criteria & Comparison Weight Assessment

Table 4 is an indication of the scale values that are used in AHP Modelling while creating a PCM.

Table 4: Scale Values

Intensity of importance	Definition
1	Equal importance
3	Moderate importance
5	Strong importance
7	Very strong or demonstrated importance
9	Extreme importance
2, 4, 6, 8	Intermediate values
1/1 to 1/9	Reciprocals values

The above-mentioned intensity of importance with their definition was used to assign respective weights during individual comparison of criteria v/s criteria.

Step 4: Normalized Weight Calculation

Firstly, from the available pairwise comparison matrix, the sum of each column was calculated. Then, each individual weight in the column were divided by the given column sum to calculate the normalized weight. The final criteria weights were calculated by averaging.

Step 5: Estimation of Consistency Ratio

CR should be less than 0.1 or else the weights assigned are inconsistent. ‘ λ ’ was calculated by dividing weighted sum with the corresponding entry. The $\lambda_{\max}$ was obtained by averaging ‘ λ ’ from all the iterations. **Equation 1** shows the formula to calculate the Consistency Index and **Equation 2** shows the formula to find the Consistency Ratio.

$$CI = \frac{\lambda_{\max} - n}{n - 1} \dots\dots 1$$

$$CR = \frac{CI}{RI} \dots\dots\dots 2$$

Since we have nine criteria, we put RI as 1.45. The final weights were obtained after this was used for ‘Weighted Overlay’.

g) Pairwise Comparison Matrix

Table 5: Pairwise Comparison Matrix

Criteria	Elevation	Slope	Aspect	LULC	Temperature	Humidity	Precipitation	Soil PH	Soil Texture
Elevation	1.000	0.500	3.000	0.333	0.500	0.500	0.333	0.250	0.250
Slope	2.000	1.000	4.000	0.500	1.000	1.000	0.500	0.333	0.333
Aspect	0.333	0.250	1.000	0.200	0.250	0.250	0.200	0.167	0.167
LULC	3.000	2.000	5.000	1.000	2.000	2.000	1.000	0.500	0.500
Temperature	2.000	1.000	4.000	0.500	1.000	1.000	0.500	0.333	0.333
Humidity	2.000	1.000	4.000	0.500	1.000	1.000	0.500	0.333	0.333
Precipitation	3.000	2.000	5.000	1.000	2.000	2.000	1.000	0.500	0.500
Soil PH	4.000	3.000	6.000	2.000	3.000	3.000	2.000	1.000	1.000
Soil Texture	4.000	3.000	6.000	2.000	3.000	3.000	2.000	1.000	1.000

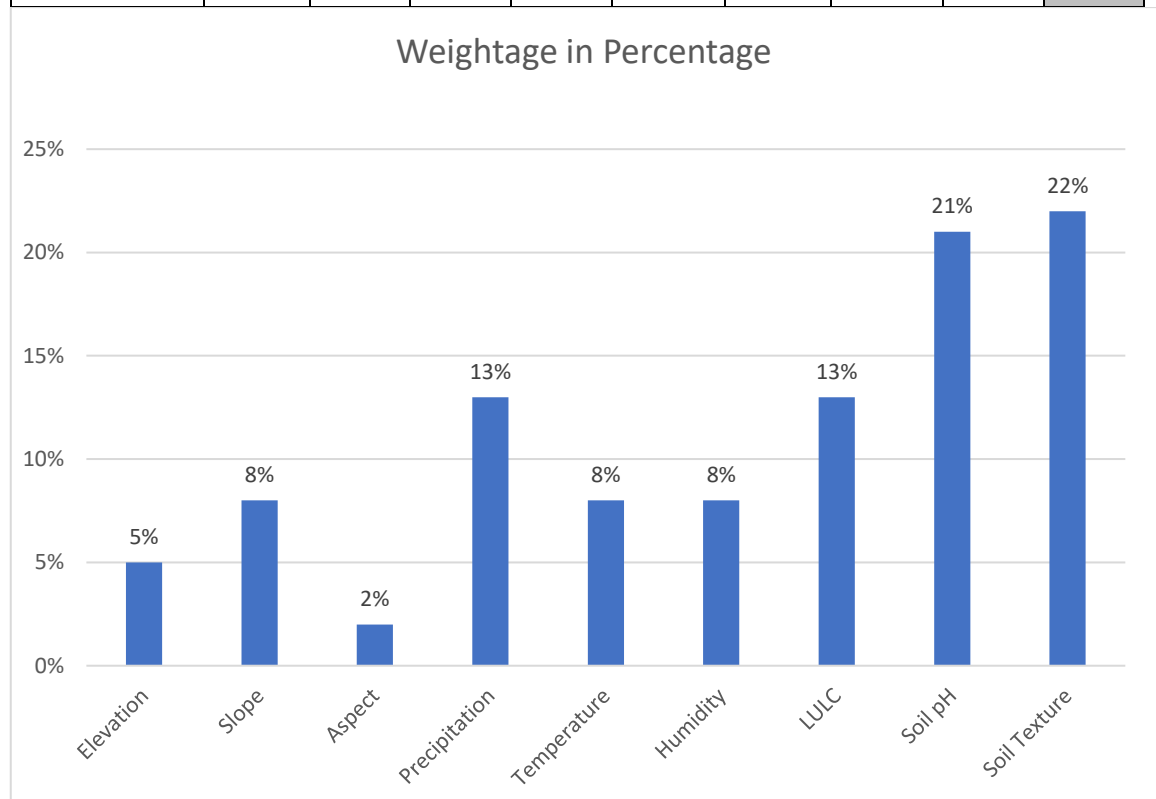


Figure 20: Criteria Weights

CHAPTER 4: PROJECT MANAGEMENT

4.1 LOGICAL FRAMEWORK

Table 6: Logical Framework

Objective	S N	Activities	Who	When (Week/s)	How	Risk	Remedies
The primary objective of the project is to evaluate the land suitability for large cardamom (Amomum subulatum) cultivation in Lamjung district using GIS and MCDM approach.	1	Planning & Literature Review	All Members	1	Group Meeting & Desk Study	Limitations of methods.	Select the suitable method for a given issue.
	2	Proposal Preparation	All Members	1,2,3	Review of the criteria for cardamom suitability	No field visit can be done.	Review from secondary data sources.
	3	Proposal Submission & Defense	All Members	4	Presentation	-	-
	4	Secondary Data Acquisition	All Members	4,5	Freely available sites	Lower accuracy of data	Trusted sites should be used
	5	AHP Modelling and Weight Calculation	Laxman Sugam	5,6	Expert Opinion for Priority Assignment. Excel for MCDA Modelling.	Priority may differ, CI value may not be met.	Adjusting the priority.
	6	GIS Analysis	Kushal Samir Samikshya	7,8	Using Different GIS tools and	Data may not provide an	Validating with historical

					functions.	accurate result	cal data
	7	Midterm Defense	All Members	9	Present ing the accom plished works	-	-
	8	Result & Interpretat ion	All Memb ers	10	Analyzing and verifying the results	Result may not align with field verificatio n	Revisitin g the weights
	9	Reporting & Document ation	All Memb ers	10,11	Using Ms word and GIS	Misleadin g errors.	Proofre ading
	10	Final Defense	All Memb ers	12	Presentation and final report submission	-	-

CHAPTER 5: BUDGET EXPENSES

Table 7: Budget Table

S.N	Particulars	Cost (in Rs)
1	Printing Cost (Proposal and Report)	1000
2	Stationeries	300
3	Secondary Data	Freely Available Used
	Total	1300

CHAPTER 6: RESULT

The goal of this project was to access the suitable places for large cardamom production in Lamjung District so that policy makers can analyze the area with high potential and help the farmers in those zones accordingly.

The land suitability analysis for large cardamom production in Lamjung District was conducted using the Analytic Hierarchy Process (AHP) integrated with GIS, incorporating nine key criteria: elevation, slope, aspect, land use/land cover, temperature, precipitation, relative humidity, soil pH, and soil texture. The final map concludes that Lamjung has 0.61% as a highly suitable zone, followed by a moderately suitable zone covering 80.37% of the total area. Also, 13.74 % area lies under the marginally suitable zone while the not suitable and restricted area accounts to 5.27% of the area.

Table 8: Area Computation of suitable zones

S.N	Suitability Zones	Area (Sq. Kilometers)	Area (ha)	AHP (% Area Coverage)
1	Highly Suitable	10.21	1021.82	0.62%
2	Moderately Suitable	1333.89	133,389.35	80.37%
3	Marginally Suitable	228.09	22,809.64	13.74%
4	Less Suitable	25.95	2,595.43	1.56%
5	Restricted	61.56	6,156.39	3.71%

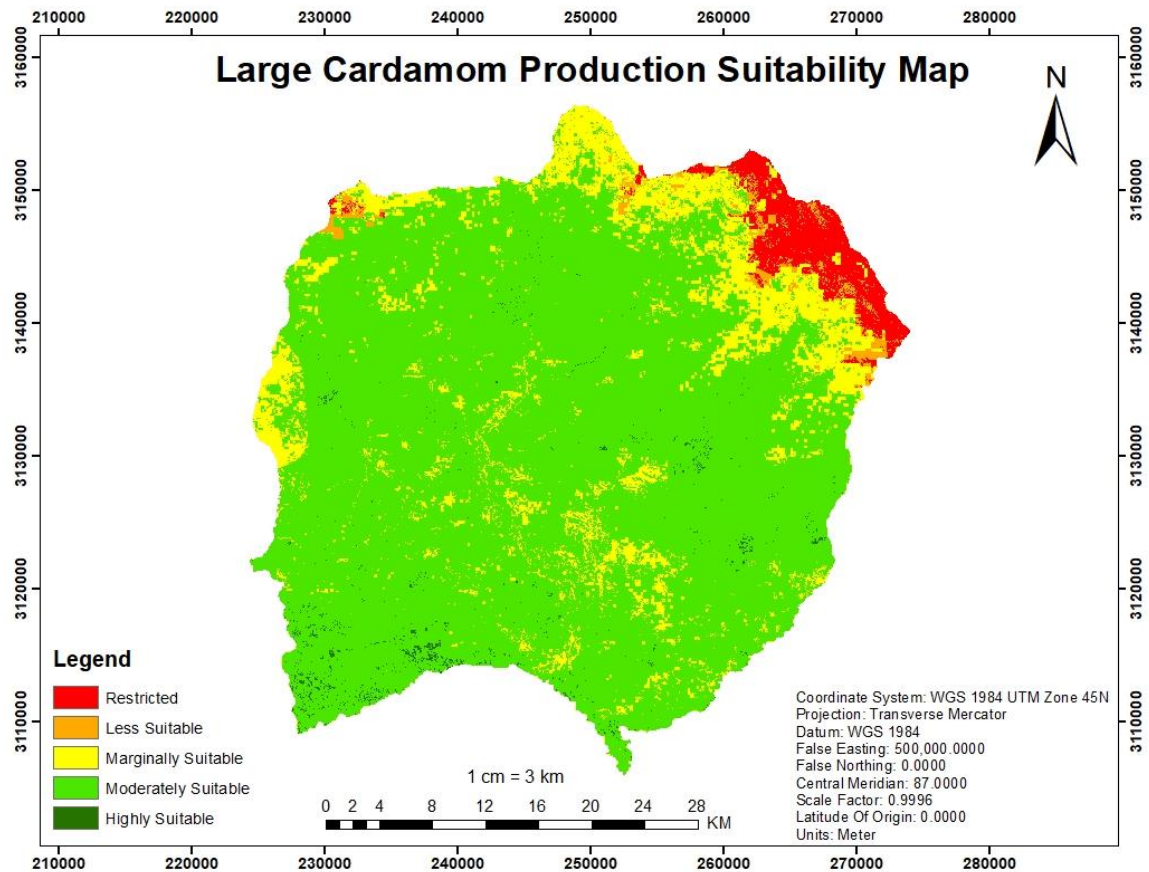


Figure 21: Large Cardamom Production Suitability Map

According to the map, moderately suitable areas (light green) cover the majority of the district, indicating that most regions possess favorable conditions for cardamom cultivation with minimal limitations. Marginally suitable zones (yellow) appear in smaller patches, particularly in the central and peripheral regions, where certain factors may not fully meet the optimal requirements. A few less suitable (orange) and restricted (red) areas are found in the northern and northeastern parts, mainly due to steep slopes, unsuitable soil, or adverse climate conditions. Only a very limited area is categorized as highly suitable (dark green), reflecting ideal conditions for cardamom growth.

6.1 FINAL MAP VERIFICATION

The Lamjung district consists of 4 municipalities and 4 rural municipalities, of which 5 of them can be considered as a pocket area for large cardamom production, namely Besishahar Municipality, Kwholasothar Rural Municipality, Dudh Pokhari Rural Municipality, Marsyangdi Rural Municipality, and Dordi Rural Municipality. The data of local areas with cardamom cultivation practice were collected through the internet, and their latitude and longitude data were plotted on the final suitability map for verification.

Table 9: Location of Validation Points

S.N	Validation Points	Longitude	Latitude
1.	Simpani -9 Marsyangdi	84°20'46"E	28°17'03"N
2.	Nalma – 5 Besishahar	84°18'01"E	28°12'31"N
3.	Gaunda – 6 Dudhpokhari	84°31'34" E	28°10'14" N
4.	Dhodeni – 5 Dudhpokhari	84°34'47" E	28°18'01" N
5	Taghring -4 Marsyangdi	84°23'21"E	28°24'48"N

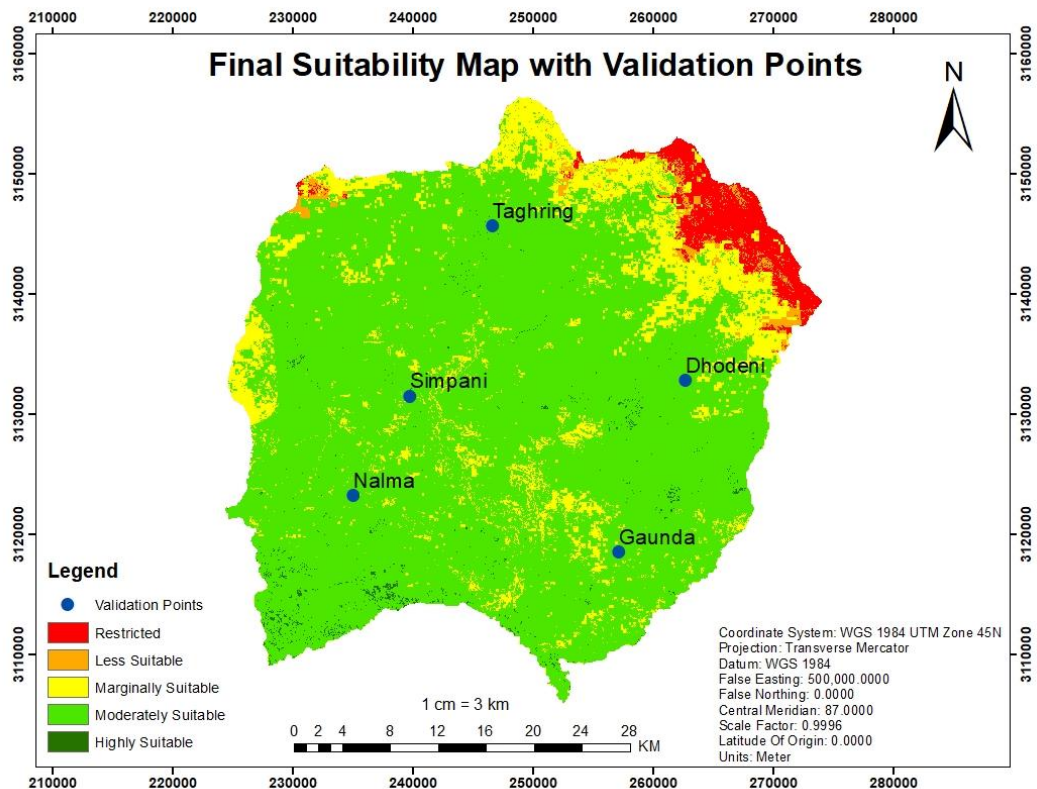


Figure 22: Final Suitability Map with Validation Points

CHAPTER 7: LIMITATION

Despite providing valuable insights, this study has several limitations.

- The analysis was based solely on secondary data sources, which may contain spatial or temporal inaccuracies, especially for climatic variables like precipitation and humidity.
- The resolution of datasets (e.g., 4–5 km for climatic layers) may not fully capture micro-climatic variations critical for cardamom cultivation.
- Additionally, the reclassification of criteria and the assignment of weights in the AHP process were based on literature review and expert judgment, which introduces subjectivity.
- Field verification was limited or absent, which restricts the validation of the model's accuracy.

CHAPTER 8: CONCLUSION

Ultimately, the project was accomplished, and the suitable zones for large cardamom cultivation were visualized using GIS with an MCDM approach. Also, the validation of the map was performed under five local areas where the cardamom is cultivated. The GIS-AHP approach proved to be a robust framework for spatially informed agricultural planning, enabling better decision-making for sustainable land use. Eventually, the findings provided valuable insights for planners, farmers, and policymakers aiming to promote economically and environmentally viable cardamom production in Lamjung District.

REFERENCES

- Acharya, M. (2019). A review on status and profitability of large cardamom production in Nepal. *International Journal of Financial, Accounting, and Management*, 1(1), 17–22. <https://doi.org/10.35912/ijfam.v1i1.53>
- Baniya, N., Böehme, M., & Baniya, S. (2009). Physical land suitability assessment for the large cardamom *amomum subulatum roxb.* Cultivation in hills of kathmandu valley. *Chinese Journal of Population Resources and Environment*, 7(4), 59–63. <https://doi.org/10.1080/10042857.2009.10684954>
- Bhattarai, T. (2016). Efficacies and Impediments in Large Cardamom Farming in Ilam, Nepal. In *Economic Journal of Development Issues* (Vol. 21, Issue 2).
- Bouamrane, A., Derdous, O., Dahri, N., Tachi, S.-E., Boutebba, K., & Bouziane, M. T. (2022). A comparison of the analytical hierarchy process and the fuzzy logic approach for flood susceptibility mapping in a semi-arid ungauged basin (Biskra basin: Algeria). *International Journal of River Basin Management*, 20(2), 203–213. <https://doi.org/10.1080/15715124.2020.1830786>
- Dawadi, P., & Pant, J. (2024). Assessment of Production and Marketing Status of Large Cardamom in Lamjung District, Nepal. *Food & Agribusiness Management*, 5(2), 93–102. <https://doi.org/10.26480/fabm.02.2024.93.102>
- Dhungana, S. M., Regmi, P. P., Dhakal, S. C., & Devkota, N. R. (2024). Suitability of Large Cardamom Farming as Business in Eastern Corridor of Nepal. *Bhartiya Krishi Anusandhan Patrika, Of*. <https://doi.org/10.18805/bkap669>
- Faisal, M., Sulisty, B., Hindarto, K. S., & Lovita, V. (2025). Evaluation of Land Suitability and Potential Development of Cardamom (*Amomum compactum* L.) in Padang Jaya Subdistrict, North Bengkulu Regency. *TERRA: Journal of Land Restoration*, 8(1), 10–19. <https://doi.org/10.31186/terra.8.1.10-19>
- Harist, M. C., Shidiq, I. P. A., Fitriani, A. H., & Santoso, A. D. (2018). A GIS-based model for a land suitability analysis of *Amomum compactum* Soland ex Maton (cardamom) in West Sumatra. *AIP Conference Proceedings*, 2021. <https://doi.org/10.1063/1.5062733>

- Malczewski, J., & Rinner, C. (2015). *Introduction to GIS-MCDA* (pp. 23–54).
https://doi.org/10.1007/978-3-540-74757-4_2
- Mohammed Abdule, A., & Kejela Woyesa, A. (2023). Physical Land Suitability Assessment for Tea Cultivation Using GIS-Based Multi-Criteria Approach at Dabus Basin of Oromia Region, Southwest Ethiopia. *American Journal of Agriculture and Forestry*. <https://doi.org/10.11648/j.ajaf.20231102.12>
- Pandey, S. K., Budhathoki, P., & Maharjan, S. K. (2023). Soil status and performance of Nepal cardamom (*Amomum subulatum* Roxb.) under different land uses. *International Journal of Horticulture and Food Science*, 5(1), 101–105.
<https://doi.org/10.33545/26631067.2023.v5.i1b.168>
- Rahim, S., Supli, A., Damiri, N., & Zaman, C. (2019). *Evaluation Tool of Land Suitability for Medicinal Plants*. 4(1), 1–6. <https://doi.org/10.22135/sje.2019.4.1.1-6>
- Shivaprasad, C. R., Korikanthimath, V. S., Gaddi, A. V, Niranjana, K. V, Venkatesh, D. H., & Krishnan, & P. (2001). Soil-site suitability evaluation for cardamom case study. In *Journal of Spices and Aromatic Crops* (Vol. 10, Issue 2).
- Timsina, K., & Pokhrel, D. (2018). *Large cardamom in Nepal: Production practice and economics, Processing and Marketing*
<https://www.researchgate.net/publication/339311342>

ANNEXES



Figure 23: Cardamom Farming in Ngadi -7 Marsyangdi



Figure 24: Farmers Harvesting Cardamom in Tarachok of Marsyangdi



Figure 25: Large Cardamom Drying and Processing



Figure 26: Cardamom ready for consumption

Column Addition:

Criteria	Elevation	Slope	Aspect	LULC	Temperature	Humidity	Precipitation	Soil PH	Soil Texture
Elevation	1.000	0.500	3.000	0.333	0.500	0.500	0.333	0.250	0.200
Slope	2.000	1.000	4.000	0.500	1.000	1.000	0.500	0.333	0.333
Aspect	0.333	0.250	1.000	0.200	0.250	0.250	0.200	0.167	0.167
LULC	3.000	2.000	5.000	1.000	2.000	2.000	1.000	0.500	0.333
Temperature	2.000	1.000	4.000	0.500	1.000	1.000	0.500	0.333	0.333
Humidity	2.000	1.000	4.000	0.500	1.000	1.000	0.500	0.333	0.500
Precipitation	3.000	2.000	5.000	1.000	2.000	2.000	1.000	0.500	0.500
Soil PH	4.000	3.000	6.000	2.000	3.000	3.000	2.000	1.000	1.000
Soil Texture	5.000	3.000	6.000	3.000	3.000	2.000	2.000	1.000	1.000
Sum	22.333	13.750	38.000	9.033	13.750	12.750	8.033	4.417	4.367

Table 10: AHP column addition

Normalized Matrix:

Criteria	Elevation	Slope	Aspect	LULC	Temperature	Humidity	Precipitation	Soil PH	Soil Texture	Priority(avg)
Elevation	0.044776	0.036364	0.078947	0.036900	0.036364	0.039216	0.041494	0.056604	0.045802	0.046273988
Slope	0.089552	0.072727	0.105263	0.055351	0.072727	0.078431	0.062241	0.075472	0.076336	0.076455568
Aspect	0.014925	0.018182	0.026316	0.022140	0.018182	0.019608	0.024896	0.037736	0.038168	0.024461435
LULC	0.134328	0.145455	0.131579	0.110701	0.145455	0.156863	0.124481	0.113208	0.076336	0.126489445
Temperature	0.089552	0.072727	0.105263	0.055351	0.072727	0.078431	0.062241	0.075472	0.076336	0.076455568
Humidity	0.089552	0.072727	0.105263	0.055351	0.072727	0.078431	0.062241	0.075472	0.114504	0.08069645
Precipitation	0.134328	0.145455	0.131579	0.110701	0.145455	0.156863	0.124481	0.113208	0.114504	0.130730327
Soil PH	0.179104	0.218182	0.157895	0.221402	0.218182	0.235294	0.248963	0.226415	0.229008	0.214938285
Soil Texture	0.223881	0.218182	0.157895	0.332103	0.218182	0.156863	0.248963	0.226415	0.229008	0.223498936

Table 11: Normalized Matrix and Priority Calculation